

# Hough Circle Transform

## **Purpose:**

Detect circles in an image using edge information and a voting technique in parameter space.

## **Concept:**

- It's based on the Hough Transform, which maps image points into a parameter space and identifies shapes by finding accumulations (votes) in that space.
- Each possible circle is represented by three parameters: circle center  $(a,b)$  and radius  $r$ .

# Algorithm

1. Preprocessing the Image: Output should be binary edge map with high-intensity pixels at edge boundaries. (Canny edge detection)
2. Edge Point Gradient Calculation
  - a. For each edge pixel (x,y), compute the image gradient direction:  $\theta = \tan^{-1} \left( \frac{dy}{dx} \right)$
  - b. The direction points toward the circle's center
3. Step 3: Voting in Accumulator Space
  - a. For each edge pixel and for each potential radius r (between minRadius and maxRadius):
  - b. Compute the potential center coordinates:  $a = x - r \cdot \cos(\theta)$  &  $b = y - r \cdot \sin(\theta)$
  - c. Increment the accumulator cell (a, b, r) by 1.
  - d. The accumulator is a 3D matrix, where each axis corresponds to a, b, and r
4. Detecting Local Maxima: After voting, each cell's value in the accumulator represents the number of votes it received - how many edge pixels agree that a circle with those parameters exists. Peaks in the accumulator space correspond to likely circles.

# Visual Representation

Circle: center  $(a, b)$  and radius  $r$

$$(x_i - a)^2 + (y_i - b)^2 = r^2$$

